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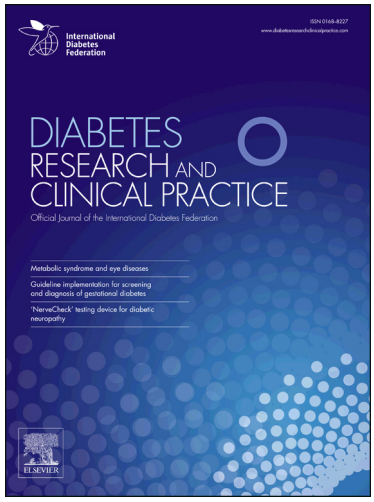

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Incidence and Risk Factors of Prediabetes and Type 2 Diabetes in a Singaporean Malay Cohort

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ABSTRACT

Aims: The aim of this work was to determine the incidence and independent risk factors of prediabetes and type 2 diabetes (T2DM) in a well-characterized cohort of Malays in Singapore.

Methods: We included 1137 participants (mean age [SD]: 55 (10) years; 53.6% female) without diabetes (DM) at baseline from the Singapore Malay Eye Study, a population-based longitudinal study with baseline (2004-2006), and follow-up (2010-2013) examinations. Prediabetes was defined as an HbA1c between 5.7% and 6.4%, with no self-reported DM history or insulin/DM medication use. T2DM was defined as a random glucose level ≥200 mg/dL or HbA1c>6.4% or use of insulin/DM medication. Age-standardized incidence was calculated as the crude 6-year cumulative incidence standardized to Singapore's Malay population census. Multivariable modified poisson regression models were utilized to determine the risk factors of incident prediabetes and T2DM.

Results: The age-standardized 6-year incidence was 11.2% (95% CI 9.5, 13.1%) for T2DM, and 20.4% (95% CI 16.4, 25.2%) for prediabetes. Hypertension, higher body mass index (BMI) and higher Hba1c levels were associated with increased risk of T2DM, while older age and higher high density lipoprotein (HDL) cholesterol were protective (all $P < 0.05$). Only higher BMI ($P=0.001$) and HbA1c ($P< 0.001$) levels were independently associated with incident prediabetes.

Conclusions: While only one in ten adult Malays developed T2DM over 6-years, one in five developed prediabetes over the same time period. Our results suggest that evidence-based interventions addressing modifiable risk factors (obesity, prediabetes, hypertension, low HDL cholesterol) are needed to delay or prevent their onset

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Key words: Incidence; Risk factor; Type 2 diabetes; Epidemiology; Obesity; Prediabetes

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INTRODUCTION

Rapid urbanization and the adoption of a more westernized lifestyle and dietary habits suggest that Asia is facing an unprecedented surge in the prevalence of type 2 diabetes (T2DM).[1] For instance, approximately 13% of the population in Singapore have diabetes (DM), second only to the United States among developed nations;[2] and the economic burden of the disease was estimated at US$787 million in 2010.[3] This has prompted the Singapore government to declare "a war on diabetes".[4] As such, high priority has been placed on research to reduce or prevent T2DM incidence and progression in these countries.

T2DM is a multifactorial disease and several key risk factors have been identified, including obesity and family history of DM.[5] Increasingly, research has focused on prediabetes, an intermediate state of hyperglycemia with blood glucose levels above normal but below DM thresholds.[6] Individuals with prediabetes are at a much higher risk of developing T2DM compared to those with normal glucose levels, with reported annual incidence rates for persons with prediabetes ranging between 5% – 15%.[7] As such, the prediabetic state represents an increasingly important focal point in the battle against T2DM particularly in Asian societies, as researchers have unanimously reported a reduction in T2DM risk after administration of drug- or lifestyle-based interventions in persons with this condition.[8, 9] It is therefore important to understand the incidence and risk factors underlying progression to a prediabetic state, as well as the factors influencing progression to T2DM at a population-based level, so that preventative interventions targeting potentially modifiable risk factors can be undertaken.

Unfortunately, evidence from population-based studies in Asian population is limited. To the best of our knowledge, in the past decade, only two Asian studies

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have been conducted to determine the incidence and independent predictors of prediabetes and T2DM development.[10, 11] More importantly, no recent data exist for populations in South-East Asia, which represent some of the most urbanized countries in the world (e.g. Malaysia, Singapore).[12] Although a 2003 study in Singapore by Wong and colleagues found T2DM incidence rates of 4.3% and 35.1% in individuals with normal and impaired glucose tolerance, respectively,[13] the study was conducted on a predominantly Chinese population; and furthermore, it is difficult to relate these findings to the present situation given the rapid changes in societal and patient profile, as a result of increasing urbanization, over the past decade. To address this knowledge gap, we examined the incidence and independent risk factors of prediabetes and T2DM incidence in an urban Singaporean Malay cohort aged 40 years or older.

RESEARCH DESIGN AND METHODS

Study population and Design

The Singapore Malay Eye Study (SiMES-1 and -2) is a population-based cohort study of Malay adults (aged 40-80 years) living in Singapore conducted between 2004-2006 and 2010-2013, respectively, which aims to assess the prevalence, incidence, progression, associated factors, and impact of major eye diseases as well as access to eye care by Asian Malays.[14] Briefly, from a list of 16,069 names living in south-western Singapore provided by the Ministry of Home Affairs, an age-stratified random sampling method was used to select 5600 names of which 4168 individuals were eligible. A total of 3280 (78.7% response rate) participants took part in SiMES-1. After 6-years, 644 (19.6%) of the 3280 subjects were ineligible to participate in SiMES-2 due to development of cognitive and/or

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severe mobility impairments, migration to other countries or having died during the intervening 6-years. Of the remaining 2636 eligible participants, 1901 (72.1% response rate) participated in SiMES-2.

All protocols followed the principles of the Declaration of Helsinki and received approval by the Singapore Eye Research Institute (SERI) Institutional Review Board. Written informed consent from participants was obtained prior to participation in the study.

Definition of Outcomes

Random glucose and HbA1c were measured from non-fasting blood samples, at the National University Hospital Reference Laboratory, which is accredited by the College of American Pathologists. HbA1c assay was carried out using HPLC cation exchange chromatography system implemented on a Biorad variant II analyzer. The assay was accredited by the National Glycoprotein Standardization Program with controls traceable to Diabetes Control and Complications Trial.[15] Following the recommendations of the American Diabetes Association (ADA), prediabetes was defined as an HbA1c ranging between 5.7% and 6.4% (39mmol/mol and 46mmol/mol),[16] with no self-reported DM history or insulin or DM medications. T2DM was defined as a random glucose level ≥200 mg/dL[16] or HbA1c >6.4% (46mmol/mol) [17] or self-reported history or insulin/DM medication use. All other individuals not falling into either of these two categories were deemed normoglycemic for the purpose of these analyses.

Assessment of Covariates

The study was conducted at the SERI research clinic and the protocol

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comprised a comprehensive, standardized examination to collect clinical and questionnaire data.[14, 18] Questionnaires to collect demographic information, sociodemographic characteristics (education, income level, occupation), lifestyle factors (smoking, alcohol, diet), self-reported family and medical history (diabetes, hypertension, thyroid diseases, stroke, cardiovascular disease, current medication use) were administered by trained interviewers, either in English or Malay according to participant preference.

Two-field fundus photos were taken using a Canon DGI non-mydriatic fundus camera and graded by trained graders at the Singapore Eye Research Institute for presence of any retinal abnormalities. Presence of retinopathy was graded by trained graders using the modified Airlie House classification system.[19]

Systolic (SBP) and diastolic (DBP) BP were measured twice using a digital BP monitor (Dinamap Pro Series DP110X-RW; GE Medical Systems Information Technologies, Inc). A third measurement was obtained if the two previous SBP readings differed by more than 10 mmHg or if DBP readings differed by more than 5 mmHg. The average value of the two closest values for each parameter were utilized in analyses. Height and weight were measured using a wall-mounted adjustable measuring scale and a calibrated scientific weight scale, respectively. BMI was calculated as weight (kg) divided by height in meters squared (kg/m^2). Aside from HbA1c and glucose, non-fasting blood samples were also collected for biochemistry analysis (total, low density lipoprotein-, and high density lipoprotein-cholesterol; high sensitivity C-reactive protein [CRP]). Hypertension was defined as having a SBP of ≥ 140 mm Hg or a DBP of ≥ 90 mm Hg or self-reported history of hypertension or anti-hypertensive medication use.

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Statistical Analyses

All analyses were performed using Stata 13.1 (StataCorp LP, College Station, TX, USA). Risk factors were analyzed either as binary traits (e.g., current smoking vs. non-smoker/past smoker) or as continuous traits (e.g., age). Cumulative six-year of 1) T2DM amongst normoglycemic and prediabetic individuals at baseline, and 2) prediabetes amongst normoglycemic individuals were calculated and further age-standardized to the 2010 Singapore population census. Due to the small number of normoglycemic individuals who developed T2DM after 6 years (N=15; **Supplementary Table 1**), we combined normoglycemic and prediabetic individuals into a single cohort at risk of developing T2DM in subsequent analyses.

To determine the factors associated with incident T2DM and prediabetes, we compared baseline characteristics between individuals who developed and did not develop each condition at the follow-up examination. Multivariable analyses were conducted using modified Poisson regression models[20] to estimate an adjusted risk ratio (RR) relating each factor to T2DM and prediabetes as separate outcomes. We adjusted for clinically important parameters and variables that were significantly different in unadjusted comparisons (**Supplementary Tables 2 and 3**), including age; gender; BMI; SBP; HbA1c; total, HDL and LDL cholesterol; low socioeconomic status (SES; monthly income<SGD$2000 and ≤6 years of education); smoking status; hypertension; prediabetes status at baseline (for incident DM analyses only); and family history of diabetes (first degree relatives, i.e. parents and siblings).

In addition, we calculated the population attributable fraction (PAF) for each risk factor significantly associated with T2DM and prediabetes incidence. PAF estimates the percentage increase/reduction in the incidence rate of a disease should the exposure be (hypothetically) completely eliminated in a given

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population.[21] Due to the possible interplay between various predictors, the PAF may add up to more or less than 100% in certain cases.[21] Continuous risk factors were dichotomized according to established guidelines to obtain interpretable PAFs: age was classified into working age (< 65 years) and retired (≥ 65 years); BMI into non-obese (≤ 27.5kg/m^2) and obese (>27.5kg/m^2) according to the world health organization (WHO) Asian obesity classifications;[22] HbA1c into normoglycemic (<5.7%/39mmol/mol) and prediabetes status (≥ 5.7%/39mmol/mol); and HDL cholesterol into recommended (>1.0 mmol/L) and below recommended (≤ 1.0mmol/L) levels.[23] As predictors of prediabetes incidence comprised solely of normoglycemic individuals at baseline (HbA1c < 5.7%/39mmol/mol), HbA1c was dichotomized using the median HbA1c value of this subpopulation (5.4%/36mmol/mol) for this particular PAF calculation. All model estimates were presented with their 95% confidence intervals and statistical significance was defined as P<0.05 in two-tailed tests.

RESULTS

Of the 3280 individuals who participated in SiMES-1, 1052 (32.1%) had diabetes and 128 (3.9%) had missing data. After excluding these individuals, 1279 (60.9%) of the remaining 2100 individuals without diabetes at SiMES-1 were re-examined in SiMES-2. This number comprised 67.3% of the total number of SiMES-2 participants (1901). Of these 1279 individuals, 142 (11.1%) had missing information, with the remaining 1137 individuals included in final analyses (**Supplementary Figure 1**).

Of the 1137 participants (609 [53.6%] females), 127 developed T2DM, giving a crude cumulative 6-year incidence of 11.2% (95% confidence interval [CI]: 9.5%,

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13.1%). No differences in cumulative incidence were found between the different age groups (P = 0.5) and the age-standardized incidence was 11.2% (9.3%, 13.4%; **Table 1**). In addition, no gender-specific difference in incidence rate was observed (P = 0.5; **Table 1**). Analyses stratified by baseline glycemic status (normoglycemia and prediabetes) revealed that just 15 (3.2%) of normoglycemic individuals developed DM after 6 years compared to 112 (16.6%) of prediabetic participants (**Supplementary Table 1**). Among the 447 normoglycemic individuals at baseline, 92 (20.6% [17.1%, 24.6%]) developed prediabetes after 6 years. The age-standardized cumulative incidence was 20.4% (16.4%, 25.2%; **Table 1**) and no age group or gender-specific differences were observed.

Table 2 shows the differences in baseline characteristics between those with incident DM and those without. Individuals who developed DM at the follow-up examination were more likely to have higher baseline BMI, SBP, CRP and lower HDL cholesterol levels; and were more likely to be hypertensive, prediabetic, and have a first-degree relative with DM. Similarly, individuals with incident prediabetes were more likely to have higher BMI, HbA1c and lower HDL cholesterol levels than those who did not develop prediabetes (**Table 3**).

Multivariable adjusted models showed that baseline presence of hypertension (risk ratio [RR]: 1.62, 95% confidence interval [CI]: 1.05, 2.51; P = 0.029), higher BMI (RR: 1.03, 95% CI: 1.00, 1.07, per unit increase; P = 0.024), and higher HbA1c levels (RR: 7.26, 95% CI: 3.07, 17.14, per % increase; P < 0.001) were predictive of incident DM, while older age (RR: 0.97, 95% CI: 0.95, 0.99, per year increase; P = 0.005), and higher HDL cholesterol levels (RR: 0.54, 95% CI: 0.30, 0.99, per unit increase; P = 0.048) were protective (**Table 4**). PAF calculated for each significantly associated variable revealed that 31% (95% CI: 5%, 49%) of the incident DM cases

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at follow-up could be attributed to presence of hypertension at baseline, 19% (95% CI: 4%, 31%) to baseline obesity (BMI> 27.5 kg/m^2), 69% (95% CI: 50%, 80%) to HbA1c between 5.7% to 6.4% (i.e. prediabetes), and 6% (0%, 12%) to lower than recommended HDL cholesterol levels (≤1.0 mmol/L) at baseline. In contrast, an 8% (95% CI: 0% to 17%) reduction in incident DM cases would have been expected if all individuals in this population were aged ≥ 65 years of age (**Table 4**).

Regarding predictors of incident prediabetes, we found that only higher BMI (RR: 1.06, 95% CI: 1.02, 1.09, per unit increase; P = 0.001) and HbA1c levels (RR: 38.8, 95% CI: 12.7, 118.1, per % increase; P < 0.001) were predictive of prediabetes development after 6 years (**Table 5**). PAF calculated for these two variables revealed that 14% (95% CI: 2%, 26%) of the incident prediabetes cases could be attributed to baseline obesity, while 44% (95% CI: 26%, 57%) could be attributed to HbA1c above the median value of 5.4% (**Table 5**).

DISCUSSION

In this longitudinal study of Singapore Malays, the 6-year age-standardized incidence of DM was 11.2%, while incident prediabetes was 20.4%, with no differences in incidence according to gender. Hypertension, higher BMI and higher HbA1c levels were independently associated with incident DM, while older age and higher HDL cholesterol levels were protective. For prediabetes incidence, higher BMI and HbA1c level were independent risk factors, while no variables appeared to be protective. PAF analyses revealed that HbA1c, hypertension and BMI were the main risk factors driving the development of these two conditions. Our results are important for researchers and policy planners to allocate resources and design

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evidence-based interventions targeting these modifiable risk factors to stem the DM epidemic in Asia.

Our finding of a 20.4% incidence for prediabetes and 11.4% for T2DM is somewhat different to two recent population-based South Asian studies on incidence and predictors of prediabetes and T2DM development. For example, the Chennai Urban Population Study (CUPS)[11] reported an 8 year incidence of 10.1% and 15.4% for prediabetes and T2DM, respectively, while the Chennai Urban Rural Epidemiology Study (CURES)[10] reported a much higher 10-year figure of 25.7% and 28.0%. Moreover, our results also contrast with those of the Singapore Impaired Glucose Tolerance follow-up study, conducted in a predominantly Chinese population more than a decade prior, where they demonstrated an 8-year prediabetes and T2DM incidence of 14.0% and 20.0%, respectively. It is unclear whether these disparities reflect true racial differences in incidence rates, or may potentially be due to dissimilarities in socio-economic status and access to medical services among other factors in the different study populations. Hence a longitudinal study, investigating possible racial differences in incidence rate and associated risk factors of prediabetes and T2DM onset in a multi-ethnic Asian population, may be warranted.

Our finding that high BMI was a risk factor for incident prediabetes and T2DM, while low HDL cholesterol levels was associated with the development for T2DM only, corroborates results from previous research.[24, 25] The underlying mechanisms of how both BMI and HDL contribute to the pathogenesis of prediabetes and T2DM are likely to be closely linked. Aside from the well-documented relationship between obesity and insulin resistance with subsequent pancreatic β-cell decompensation in the pathogenesis of T2DM,[24] recent studies

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have identified obesity-induced T2DM pathogenic pathways including increased levels of proinflammatory cytokines, deranged fatty acid metabolism, and cellular process e.g. mitochondrial dysfunction and endoplasmic reticulum stress.[26] Experimental studies have also implicated increased HDL cholesterol levels in upregulating β-cell insulin secretion;[27] conventional wisdom would therefore suggest that low levels of HDL cholesterol, in combination with obesity, would significantly increase the risk of incident prediabetes and T2DM. However, while low HDL cholesterol levels were associated with incident T2DM in our study sample, we are unsure as to why it did not significantly contribute to the development of prediabetes. Further studies to replicate our observation and better elucidate the underlying mechanisms are needed.

Presence of hypertension at baseline was also associated with incident T2DM, with 31% of cases being attributed to this variable. This finding is in-line with that of other cohort studies in both Caucasian[28] and Asian[29] populations, and several mechanisms have been proposed to underlie this relationship, although a direct causal link has yet to be established. An altered endothelial permeability and diminished blood flow to the peripheries (vascular resistance) as a result of hypertension may limit insulin delivery and promote insulin resistance in metabolically active tissues.[30] The oxidative stress associated with high blood pressure has also been postulated to play a critical role in pancreatic β-cell dysfunction.[31] In addition, cytokines related to oxidative stress, such as interleukin-1, interleukin-6, and tumor necrosis factor-α, has been theorized to be able to potentially modify glucose metabolism.[32] The above combination of systemic vascular resistance, oxidative stress and low-grade chronic inflammation therefore could conceivably increase insulin resistance and lead to the development of T2DM.

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We observed that high HbA1c levels were strongly and independently associated with risk of incident prediabetes and T2DM. In particular, the prediabetic state has been observed to carry a significantly increased risk of transitioning to actual T2DM,[6, 7] and this holds true in our sample of Singaporean Malays: the age-standardized cumulative incidence of T2DM was over 5 times higher in individuals with prediabetes at baseline compared to those without and approximately two-thirds of the incident T2DM risk could be attributed to having baseline prediabetes. While the role of HbA1c in development of prediabetes is less well-researched, higher HbA1c levels being the major contributing factor to the development of this condition is plausible, as prediabetes and T2DM are essentially chronological representations of the same disease albeit at different stages. Given the enormous economic burden associated with DM, public health interventions such as exercise or pharmacological therapy[8, 9] that have been shown to slow, prevent or even reverse the development of prediabetes are strongly needed to halt the T2DM epidemic sweeping through Asia.

Surprisingly, a family history of DM in first degree relatives (parents/siblings) was not a significantly risk factor in the development of prediabetes or T2DM, which contrasts with established literature showing the opposite.[33, 34] We hypothesize that those with a positive family history may have a higher HbA1c and BMI which may confound the relationship between a positive family history and prediabetes and T2DM. Indeed, we found a significant association between a positive family history of diabetes, with BMI (P = 0.032) and HbA1c (P = 0.055, data not shown). In addition, it is also possible that participants who had a positive family history of diabetes may have taken preventative measures to reduce T2DM risk, such as increasing their

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daily physical activity and controlling their dietary intake of carbohydrates, which we were unable to assess in our study as these variables were not collected.

Interestingly, older age was found to be protective for incident T2DM (but not prediabetes) in our study, which contrasts with findings in Asia[10] and elsewhere[35, 36] where older age was a risk factor for both of these conditions. This discrepancy may be due to survival bias, i.e. older persons who would have developed prediabetes or T2DM at follow-up may have passed away or simply not attended due to poor health. Indeed, we showed that those who did not participate in the SiMES-2 examination were older at baseline; more likely to be prediabetic and hypertensive; and have higher SBP and DBP (**Supplementary Table 2**).

Strengths of this study include a longitudinal population-based study design, allowing inferences on the temporality of the observed relationship between exposure and outcome, a large sample size, standardized methodology and an extensive clinical evaluation. Limitations include the fact that prediabetes status was categorized through HbA1c levels and not by the traditional fasting and 2-hour post-prandial glucose measurements.[37] However, HbA1c and traditional fasting plus 2-hour glucose measurements provide comparable results in terms of predicting the incidence of T2DM and cardiovascular diseases from prediabetes.[38, 39] Selection/survival bias may also be present due to differences in key characteristics between participants and non-participants in SiMES-2, hence our findings should be interpreted with caution. In addition, we utilized anthropometric measures (i.e. BMI) as a surrogate for obesity in this study; however, this parameter is unable to distinguish between visceral and subcutaneous fats. An assessment of regional adiposity, e.g. using dual x-ray absorptiometry (DXA), may better tease out the nuances in the associations between obesity and prediabetes/T2DM.[40] Lastly, our

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study was done in ethnic Malays and may not be applicable to Asians of other ethnicities (e.g. Chinese, Indians).

In conclusion, approximately 1 in 5 and 1 in 10 adult Malays aged 40 years and over developed prediabetes and T2DM, respectively, over a 6 year period. Our findings have huge societal, patient-centered and economic implications for Asia given the expected exponential increase in T2DM cases in the coming decades.[1] Public health screening, together with clinical, lifestyle, and behavioral intervention programs focused on addressing the modifiable risk factors of prediabetes and T2DM (obesity, prediabetes, hypertension and low levels of HDL cholesterol), are strongly warranted to reduce the incidence of these two conditions in the ethnic Malay population.

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Table 1. Cumulative 6-year progression to prediabetes and diabetes in ethnic Malays (N=1137)

	Prediabetes			Diabetes		
Parameter	**No. at risk**	**incidence (%)**	**95% CI**	**No. at risk**	**incidence (%)**	**95% CI**
Age group (years)						
40-49	199	46 (23.1)	(17.7, 29.5)	399	47 (11.8)	(9, 15.3)
50-59	147	29 (19.7)	(14.0, 27.0)	407	45 (11.1)	(8.3, 14.5)
60-69	65	11 (16.9)	(9.5, 28.4)	203	23 (11.3)	(7.6, 16.5)
> 70	36	6 (16.7)	(7.4, 33.4)	128	12 (9.4)	(5.4, 15.9)
		P-trend = 0.2			*P-trend = 0.5*	
Crude	447	92 (20.6)	(17.1, 24.6)	1137	127 (11.2)	(9.5, 13.1)
Age-standardized*		20.4	(16.4, 25.2)		11.2	(9.3 to 13.4)
Gender						
Male	170	36 (21.1)	(15.6, 28.0)	528	55 (10.4)	(8.1, 13.3)
Female	277	56 (20.2)	(15.9, 25.4)	609	72 (11.8)	(9.5, 14.6)
		P = 0.8[†]			*P = 0.5*[†]	

Abbreviations: CI = confidence interval

* Age-standardized rates (95% CI) compared with the 2010 Singapore population.

[†]: P value for difference between gender

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Table 2. Comparison of baseline characteristics of individuals with incident diabetes versus those who remained non-diabetic at follow-up (N=1137)

Characteristics	Non-DM (N = 1010)	Incident DM (N = 127)	P-value
	Mean (SD)	Mean (SD)	
Age, years	55.2 (9.9)	54.4 (9.7)	0.4
BMI	25.8 (4.5)	28.5 (5.3)	**<0.001**
Systolic BP, mmHg	140.4 (21.7)	144.6 (20.4)	**0.03**
Diastolic BP, mmHg	78.9 (10.7)	80.0 (11.5)	0.3
C-reactive protein, mg/L	3.4 (7.0)	5.3 (12.9)	**0.009**
Total cholesterol level, mmol/L	5.6 (1.1)	5.7 (1.2)	0.4
HDL cholesterol level, mmol/L	1.4 (0.3)	1.3 (0.3)	**0.002**
LDL cholesterol, mmol/L	3.5 (0.9)	3.6 (1.0)	0.5
HbA1c, %	5.7 (0.4)	6.0 (0.3)	**<0.001**
Characteristics	**Non-DM**	**Incident DM**	**P-value**
	%	%	
Gender, Female	53.2	56.7	0.5
Hypertension, yes	56.1	74.0	**<0.001**
Low socio-economic status*, yes	8.5	11.0	0.3
Family history of diabetes, yes	29.1	38.6	**0.03**
Current smoker, yes	20.2	13.4	0.07
Prediabetes, yes	55.7	88.2	**<0.001**
Any retinopathy, yes	5.8	5.5	0.9

*Low socio-economic status: ≤ 6 years education, individual monthly income <SG$2000, living in 1 or 2 room flats

DM: Diabetes mellitus, HDL: High density lipoprotein, LDL: Low density lipoprotein

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Table 3. Comparison of baseline characteristics of individuals with incident prediabetes versus those who remained normoglycemic at follow-up

Characteristics	Normoglycemic (N = 355)	Incident Prediabetes (N = 92)	P-value
	Mean (SD)	Mean (SD)	
Age, years	53.4 (9.8)	52.2 (9.4)	0.3
BMI	24.7 (4.1)	26.7 (5.3)	**<0.001**
Systolic BP, mmHg	137.3 (22.3)	137.4 (21.1)	0.9
Diastolic BP, mmHg	78.1 (10.8)	77.9 (11.6)	0.9
C-reactive protein, mg/L	2.6 (5.7)	2.8 (3.1)	0.7
Total cholesterol level, mmol/L	5.4 (0.9)	5.3 (1.0)	0.5
HDL cholesterol level, mmol/L	1.4 (0.3)	1.3 (0.3)	**0.04**
LDL cholesterol, mmol/L	3.3 (0.8)	3.4 (0.9)	0.9
HbA1c, %	5.3 (0.2)	5.4 (0.1)	**<0.001**
Characteristics	**No Change**	**Incident Prediabetes**	**P-value**
	%	%	
Gender, Female	62.2	60.8	0.8
Hypertension, yes	46.4	46.7	0.9
Low socio-economic status*, yes	10.1	10.8	0.8
Family history of diabetes, yes	26.2	30.4	0.4
Current smoker, yes	19.4	17.3	0.7
Any retinopathy, yes	4.7	5.4	0.8

*Low socio-economic status: ≤ 6 years education, individual monthly income <SG$2000, living in 1 or 2 room flats

HDL: High density lipoprotein, LDL: Low density lipoprotein

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Table 4. Multivariable* adjusted risk factors of 6-year incident type 2 diabetes in ethnic Malays

Parameters	Incident T2DM (N = 1137)		
	RR (95% CI)	P-value	PAF (95% CI)
Age, per year increase	**0.97 (0.95, 0.99)**	**0.005**	-
<65 years	Reference	-	-
≥ 65 years	0.65 (0.41, 1.05)	0.077	-0.08 (-0.17, 0.00)
Gender, female	1.13 (0.78, 1.63)	0.5	-
Hypertension, yes	**1.62 (1.05, 2.51)**	**0.029**	0.31 (0.05, 0.49)
Low socio-economic status, yes	1.55 (0.92, 2.61)	0.1	-
Family history of diabetes, yes	1.11 (0.80, 1.55)	0.5	-
Current smoker, yes	0.67 (0.40, 1.11)	0.1	-
BMI, per unit increase	**1.03 (1.00, 1.07)**	**0.024**	
Not obese (≤ 27.5kg/m2)	Reference	-	-
Obese (>27.5kg/m2)	**1.57 (1.12, 2.19)**	**0.008**	0.19 (0.04, 0.31)
Systolic BP, per mmHg increase	1 (0.99, 1.01)	0.9	-
HbA1c, per % increase	**7.26 (3.07, 17.14)**	**<0.001**	-
< 5.7%/39mmol/mol	Reference	-	-
≥ 5.7%/39mmol/mol	**4.54 (2.65, 7.78)**	**<0.001**	0.69 (0.50, 0.80)
C-reactive protein, per unit increase	1.01 (0.99, 1.02)	0.9	-
Total cholesterol, per unit increase	1.04 (0.89, 1.21)	0.5	-
HDL cholesterol, per unit increase	**0.54 (0.30, 0.99)**	**0.048**	-
>1.0 mmol/L	Reference	**-**	-
≤ 1.0mmol/L	**1.69 (1.04, 2.76)**	**0.034**	0.06 (0.00, 0.12)

*includes all variables in analyses

DM: Diabetes, BMI: body mass index, BP: Blood pressure, HDL: High density lipoprotein, LDL: Low density lipoprotein; PAF: population attributable fraction

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Table 5. Multivariable* adjusted risk factors of 6-year progression to prediabetes in baseline normoglycemic ethnic Malays (N = 447)

Parameters	Progression to prediabetes		
	RR (95% CI)	P-value	PAF (95% CI)
Age, per year increase	0.98 (0.96, 1.01)	0.2	-
Gender, female	0.91 (0.56, 1.48)	0.7	-
Hypertension, yes	0.94 (0.56, 1.58)	0.8	-
Low socio-sconomic status, yes	1.21 (0.7, 2.07)	0.5	-
Family history of diabetes, yes	1.12 (0.76, 1.65)	0.6	-
Current smoker, yes	0.91 (0.52, 1.59)	0.7	-
BMI, per unit increase	**1.06 (1.02, 1.09)**	**0.001**	**-**
Not obese (≤27.5kg/m2)	Reference	-	-
Obese (>27.5kg/m2)	1.61 (1.11, 2.36)	0.013	0.14 (0.02, 0.26)
Systolic BP, per mmHg increase	1 (0.99, 1.02)	0.6	-
HbA1c, per % increase	**38.8 (12.7, 118.1)**	**<0.001**	**-**
Lower Quantile (≤ 5.4%/36mmol/mol)	Reference	-	-
Upper Quantile (> 5.4%/36mmol/mol)	**2.77 (1.87, 4.12)**	**<0.001**	0.44 (0.26, 0.57)
C-reactive protein, per unit increase	1 (0.97, 1.02)	0.7	-
Total cholesterol, per unit increase	0.9 (0.74, 1.11)	0.3	-
HDL cholesterol, per unit increase	0.85 (0.43, 1.68)	0.6	-

*includes all variables in analyses

BMI: body mass index, BP: Blood pressure, HDL: High density lipoprotein, LDL: Low density lipoprotein; population attributable fraction (PAF)

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Highlights

- One in ten Singaporean Malay adults developed type 2 diabetes over 6-years
- Over the same time period, one in five developed prediabetes
- Most risk factors, e.g. Hypertension, higher BMI and HbA1c levels, were modifiable
- Intervention programs addressing these factors are warranted

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